Longest Palindromic Substring

AlgoExpert · Medium · Strings

The problem

- Expand around every centre (odd and even).

Intuition

- Every palindrome has a centre; expand around each of the 2n minus 1 centres.

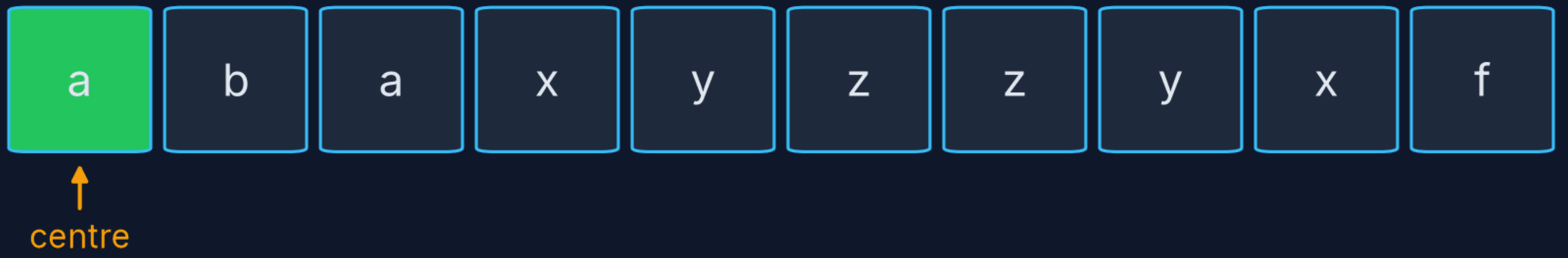
Brute force vs optimal

- Check every substring → $O(n^3)$
- DP table → $O(n^2)$ time and space
- Expand around $2n - 1$ centres → $O(n^2)$ time, $O(1)$ space

Approach

- 1. For each index, expand odd and even centres
- 2. Track the longest span
- 3. Return the substring

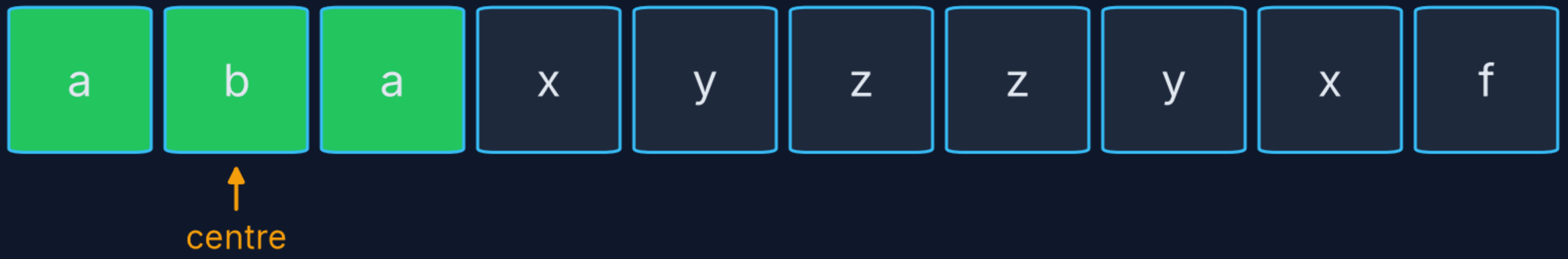
Worked example




```
odd centre at 0 → 'a'
```

New best: 'a'

Worked example




```
odd centre at 1 → 'aba'
```

New best: 'aba'

Worked example




```
even centre at 5 → 'xyzzyx'
```

New best: 'xyzzyx'

Java solution

```java
public class LongestPalindromicSubstring {
    public static String longestPalindromicSubstring(String s) {
        int bs = 0, be = 0;
        for (int i = 0; i < s.length(); i++)
            for (int j = i; j <= i + 1; j++) {
                int l = i, r = j;
                while (l >= 0 && r < s.length() && s.charAt(l) == s.charAt(r)) { l--; r++; }
                if (r - l - 1 > be - bs) { bs = l + 1; be = r; }
            }
        return s.substring(bs, be);
    }
}
```

Complexity

- Time: O(n^2)
- Space: O(n) output

Edge cases & pitfalls

- Even-length palindromes need gap centres
- Single char string
- Manacher's for O(n)

Interview tips & follow-ups

- Mention Manacher's algorithm

Recap

- Every palindrome has a centre; expand around each of the 2n minus 1 centres.
- Complexity: O(n^2)